

Nikita Kirnosov

Senior Staff Software Engineer / Manager

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Skills

Technical Leadership, AI/ML Platforms Design, High Load Systems, Recommenders.

Languages C/C++, Python, Java, Go, SQL.

Experience

2024 Apr – **Senior Staff Software Engineer / Senior Engineering Manager, Google - Search.**

Present Created Search Discovery Foundations team composed of Updates Infrastructure, ML Platform, and AI Platform sub-teams.

- Grew the team by 20 engineers in 6 months.
- Landed real-time (15 minute fresh) transformer architecture recommendation models training and experimentation (+32% engagement).
- Built GenAI infrastructure in collaboration with Gemini and launched:
 - Topic updates generation: users receive meaningful updates on the followed queries (16% re-engagement increase).
 - [Samsung Now Bar](#): users follow the games with real-time commentary generated from the UGC.
 - [Daily Listen](#): daily AI-generated podcasts on the news personalized to your interests.
- Enhanced serving infrastructure to support live activities with industry-leading latencies.

2021 Apr – **Staff Software Engineer / Engineering Manager, Google - Search.**

2024 Apr Lead a Recommendations team powering notifications, promos, and suggestions to deliver >20% of inorganic Search growth in 2023. Established and led cross-org collaboration of 27 engineers from Google Research, Core Infrastructure, and Android, enabling 83 client team SWEs and PM work by changed user-content matching approach and adopting recent AI advances.

- Reduced new update product idea-to-MVP time from a month to 2 hours and time to launch from a quarter to 2 weeks.
- Created a full cycle ML/AI ecosystem (indexing, generation, matching, training, serving).
 - Augmented heuristic-based content-user matching rules with ML based content retrieval, improving user engagement by 24%.
 - Created scalable architecture supporting various retrieval strategies (item2item, ScaNN, etc.), yielding 10% engagement increase in the first quarter.
- Introduced on-device ML to a stack, improving notification triggering quality by 7%.

2018 Apr – **Senior Software Engineer, Google - Search.**

2021 Apr Envisioned and lead the team implementing ML training and serving ecosystem for Search's News and Content Discovery.

- Designed and lead the implementation of unified and standardized ML infrastructure (C++) for Discover, Notifications, Suggest and user modeling teams featuring:
 - zero training/serving skew near real time training data generation system, reducing the feature engineering work by 70%.
 - dynamic triggering rate optimization based on recent and long-term user activity pattern.
- Scaled an existing inference serving infrastructure to serve 24x more inferences using the same resources.
- Restructured Discover Search training process to support training on TPU to achieve 40x speedup (Python).
- Created user opt-out detection system proactively stopping bad experiments (C++).

2016 Apr – **Software Engineer, Google - YouTube.**

2018 Apr Built storage abstraction microservices framework with no-downtime migration support (C++) and improved YT Public Data API performance, usability, and security (Python).

- Designed and implemented storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type (C++).
- Migrated video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase (C++).

2015 **Fellow, Insight Data Science.**

Sep–Oct Built [CityGravity.info](#), a trip schedule optimization app to see the greatest number of attractions.

2010 Aug – **Graduate Researcher, University of Arizona, Tucson, AZ.**

2016 Jan Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results are published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.

2010 **Intern, Joint Institute of Nuclear Research, Dubna, Russia.**

Jan–May Contributed to muon detector for CMS software development by assessing signal processing code and vectorizing it.

Education

2010–2015 **Ph.D., University of Arizona, Tucson, AZ – Physics.**

2005–2010 **B.S. (Hons.), Saratov State University, Saratov, Russia – Physics.**